Portfolio demo. The shop and every number come from a **fictional dataset** generated by `make_data.py`. The analysis itself is real and reproducible.

Does the one-page checkout sell more?

Northwind Outfitters (fictional) · A = current 3-step checkout, B = new one-page checkout · 14 days · 41,800 visitors

Recommendation: ship B. It lifts conversion by about +15% (+4% to +26%, p = 0.006). Plan for a long-run lift closer to +13%: the first days were inflated by a novelty effect. Keep watching order value, which may dip slightly.

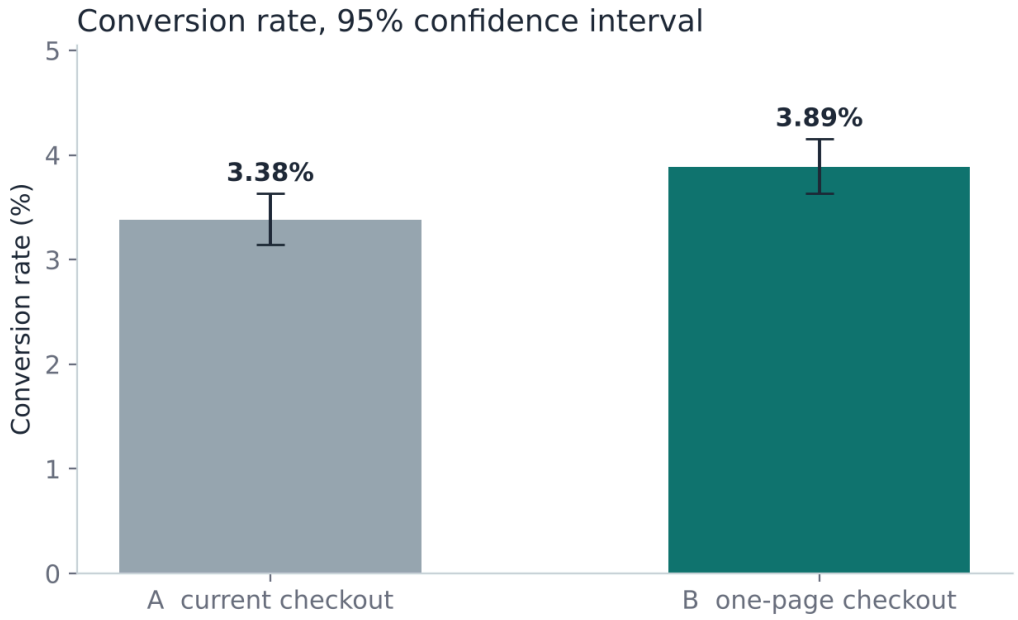
+15%	**+12%**	**−€1.82**
conversion rate	revenue per visitor	average order value
3.38% → 3.89%	€2.48 → €2.79 (borderline)	not conclusive

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1. Is the test itself trustworthy?

- **Traffic split:** 20,925 vs 20,875 visitors. A sample-ratio check against 50/50 gives p = 0.81, so no sign of a broken randomiser.
- **Sensitivity:** with this sample, the test could reliably detect (80% power) a lift of about +15% or more. The observed lift is right at that size, so the result is real but the exact number is uncertain.

2. Conversion

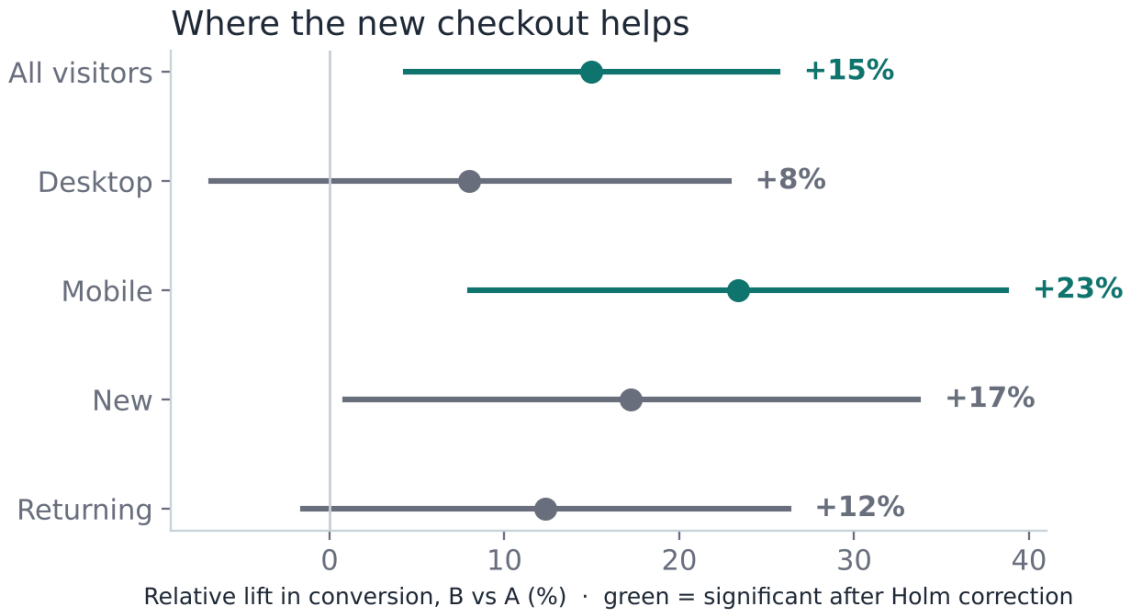


B converts 3.89% of visitors vs 3.38% for A: +0.51 percentage points (95% CI +0.15 to +0.87). Two-proportion z-test p = 0.006.

3. Money, not just clicks

- **Revenue per visitor:** €2.48 → €2.79. Bootstrap 95% CI for the difference € +0.01 to €+0.61; Welch t-test p = 0.054. Likely positive, but not as solid as the conversion result.
- **Average order value:** €73.53 → €71.71 (CI €-6.28 to €+2.62). Buyers in B may spend a little less per order, a plausible side effect of showing fewer add-ons. Worth monitoring after launch.

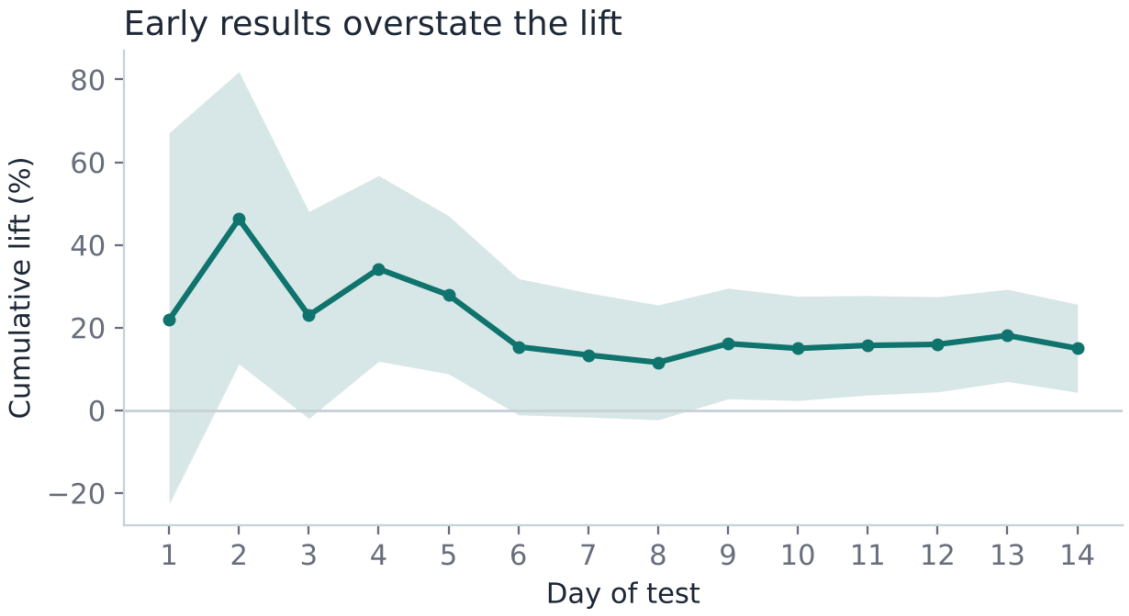
4. Who benefits?



Segment	Visitors	A	B	Lift (95% CI)	p (Holm)
Desktop	15,090	4.62%	4.99%	+8% (-7% to +23%)	0.289
Mobile	26,710	2.66%	3.28%	+23% (+8% to +39%)	0.011
New	24,173	2.51%	2.94%	+17% (+1% to +34%)	0.116
Returning	17,627	4.59%	5.15%	+12% (-2% to +26%)	0.161

The gain is clearest on **mobile** (+23%, Holm-adjusted p = 0.011). On desktop the lift is +8% and not significant. A formal interaction test (logistic regression) gives p = 0.18, so we **cannot** claim the effect truly differs by device. Segment results are corrected for multiple comparisons (Holm).

5. Watch out for early results



On day 2 the cumulative lift read +46% with p = 0.010. Stopping there would have overstated the gain about threefold. Days 1–3 show +23%; days 4–14 show +13%. That pattern fits a novelty effect, so the steady-state estimate is the safer number for forecasting.

Methods

Sample-ratio chi-square test; two-proportion z-test with Newcombe CI; bootstrap (4,000 resamples) and Welch t-test for revenue per visitor; bootstrap CI and Mann-Whitney U for order value; per-segment z-tests with Holm correction; logistic regression `converted ~ variant * device + visitor_type` for the interaction; power via Cohen's h at α = 0.05. Python: pandas, SciPy, statsmodels, matplotlib.